SG No. 64 D_{2h}^{18} $Cmce$ [orthorhombic]

* generator : $\{2_{001}|0\frac{1}{2}\frac{1}{2}\}$, $\{2_{010}|0\frac{1}{2}\frac{1}{2}\}$, $\{-1|0\}$
* symmetry operation $+[0,\,0,\,0]$, $+\left[\frac{1}{2},\,\frac{1}{2},\,0\right]$

Table 1: Symmetry operations for 3d polar vector.

No.	tag	matrix (polar)	det
1	$\{1\|0\}$	$\begin{bmatrix}1&0&0&0\\0&1&0&0\\0&0&1&0\end{bmatrix}$	1
2	$\{2_{001}\|0\frac{1}{2}\frac{1}{2}\}$	$\begin{bmatrix}-1&0&0&0\\0&-1&0&\frac{1}{2}\\0&0&1&\frac{1}{2}\end{bmatrix}$	1
3	$\{2_{010}\|0\frac{1}{2}\frac{1}{2}\}$	$\begin{bmatrix}-1&0&0&0\\0&1&0&\frac{1}{2}\\0&0&-1&\frac{1}{2}\end{bmatrix}$	1
4	$\{2_{100}\|0\}$	$\begin{bmatrix}1&0&0&0\\0&-1&0&0\\0&0&-1&0\end{bmatrix}$	1
5	$\{-1\|0\}$	$\begin{bmatrix}-1&0&0&0\\0&-1&0&0\\0&0&-1&0\end{bmatrix}$	−1
6	$\{\mathrm{m}_{001}\|0\frac{1}{2}\frac{1}{2}\}$	$\begin{bmatrix}1&0&0&0\\0&1&0&\frac{1}{2}\\0&0&-1&\frac{1}{2}\end{bmatrix}$	−1
7	$\{\mathrm{m}_{010}\|0\frac{1}{2}\frac{1}{2}\}$	$\begin{bmatrix}1&0&0&0\\0&-1&0&\frac{1}{2}\\0&0&1&\frac{1}{2}\end{bmatrix}$	−1
8	$\{\mathrm{m}_{100}\|0\}$	$\begin{bmatrix}-1&0&0&0\\0&1&0&0\\0&0&1&0\end{bmatrix}$	−1